

Veritas-Medica -- Engineering Work Breakdown

Date: 2026-08-03 **Companion to:** [MARKET_READINESS_ROADMAP.md](#) (the *what* and *why*) -- this is the *how*, at a level an engineer can pick up.

Every item below was checked against the current tree. Where the roadmap assumed something exists, and it does not, that is called out -- three of those materially change the estimates.

Effort is in person-days for one engineer familiar with the codebase. Multiply by ~1.6 for someone new to it.

Findings that revise the roadmap's assumptions

#	Assumption	Reality	Effect
A	"Accuracy evaluation pack" is a reporting exercise	No evaluation harness exists. docs/evidence-pack-e0.md is a 73-line document; there is no evals/ directory, no labelled dataset, no scoring code	E1 grows from ~3d to ~12d and needs clinical input
B	DSR module gives erasure to build on	dsr.service.ts only records requests (status: 'pending') -- createErase writes a row, it does not delete anything, in Postgres or Neo4j	E8 is larger than "add Neo4j parity"; there is no working erasure anywhere
C	Checklist persistence is a small add	Zero backend -- 0 references to checklist in apps/core	Needs migration + service + controller + client, not just a hook change

None of these are blockers. All three were simply scoped optimistically because the evaluation looked at *architecture*, not at whether each capability had a working implementation behind it.

NOW -- 6 weeks

E2 -- Port orchestrator to [ModelProvider](#) -- 4 days (*highest RICE, 8.1*)

Exists: [apps/qa/src/qa/model_client.py](#) has the target pattern -- [ModelProvider](#) Protocol ([complete\(\)](#) / [version\(\)](#)), [StubModelProvider](#), [LocalModelProvider](#) over OpenAI-compatible [/chat/completions](#), PHI guard wired, [get_model\(\)](#) factory.

Missing: [services/orchestrator/deepseek_client.py](#) has none of it -- [DEFAULT_BASE_URL](#) = ["https://api.deepseek.com"](#) hardcoded, no Protocol, no provider switch, no guard. All five agents route through it.

Build:

- [services/orchestrator/model_provider.py](#) -- port the Protocol + both providers from [apps/qa](#). Do **not** re-invent; keep the interface identical so the two services stay comparable.
- [pyproject.toml](#) -- add [phi-guard](#) path dependency (pattern already used by [services/nphies-engine](#)); bump [requires-python](#) to [>=3.12](#) to match.

1 Refactor `_chat_completion()` to delegate to the provider. `format_agent_prose` and `generate_soap_note` signatures stay unchanged -- nothing downstream moves.

1 Env: `ORCHESTRATOR_MODEL_PROVIDER=stub|local|deepseek`,
`MODEL_ENDPOINT_URL`, `MODEL_NAME`, reusing existing names.

1 Tests: provider selection; PHI-guard refusal for an external endpoint under
`PHI_EGRESS_POLICY=block`; existing 59 orchestrator tests still green.

Acceptance: a self-hosted model swap is a `.env` change with no code edit, demonstrated live. Audit **H-1 closes**.

Watch: `deepseek_client.py` currently raises when `DEEPSEEK_API_KEY` is absent. Preserve the Sprint-10 degradation fix -- the provider must fail in a way `receptionist_agent.generate_care_instructions()` can still catch, or the post-care package regresses to 500s.

E4 -- Finish UI wiring -- 6 days

Three separate gaps, different sizes.

E4a -- Agent action cards (M-1) -- 2 days. `SullyContext.runAgentAction()` appends a message and returns; eight buttons across five tabs do nothing. Split them:

Action	Disposition
"Regenerate SOAP note"	Wire -> E4b's endpoint
"Summarise prior encounters"	Wire -> existing <code>/narrative</code>
"Show related results"	Wire -> existing <code>/observations</code>
"Submit Pre-Auth"	Wire -> existing <code>submitPreAuth</code>
"Adjust Dosage", "Check formulary tier", "Fix code mismatch", "Book follow-up", "Send visit summary"	Mark "Pending integration" using the <code>ReceptionistTab</code> pattern from Phase 1

Do not invent backends to make buttons light up. The disabled-with-explanation treatment is already established and is the honest option.

E4b -- SOAP generation (H-3b) -- 3 days. `generate_soap_note(transcript)` exists in the orchestrator and is never called. Needs: a core proxy route (`POST patients/:id/ai-team/soap` -- same scope/RBAC/ audit shape as the Phase 2 routes), `api.aiTeam.generateSoap()`, and `SullyContext` calling it when live transcript lines land, replacing `SOAP_STAGES`. Keep the canned stages under `dictationMode === "demo"`.

E4c -- Checklist persistence (M-7) -- 1 day. See finding **C** -- this is a new migration (`app.encounter_checklist`), service, controller and client method, not a hook tweak.

E1 -- Clinical accuracy evaluation pack -- 12 days (revised, finding A)

Exists: nothing executable. `docs/evidence-pack-e0.md` describes a prior gate but there is no harness, dataset or scoring code.

Build:

- `evals/` -- dataset format (JSONL: input, expected, rationale) and a runner emitting precision / recall / F1 per task.
- Labelled sets** (needs a clinical coder -- the long pole, not the code):

- ICD-10 coding suggestions: ≥ 200 conditions with correct codes
 - SBS order coding: ≥ 150 orders
 - NPHIES necessity: ≥ 100 diagnosisxorder pairs with known verdicts
- 1 **Scribe WER**: ≥ 30 recorded consultations (Arabic + English), reference transcripts, WER per language. Synthetic/consented audio only.
 - 1 **NSCRE regression set**: the DDI and renal-dose scenarios as fixed cases, so a reference-data change cannot silently alter clinical output.
 - 1 **just eval** + CI job; a one-page report artefact.

Acceptance: a published one-pager with real numbers. **Publish them even if unflattering** -- measured weakness beats unmeasured confidence in front of a CMO.

Sequencing note: start dataset labelling in week 1. The code is ~4 days; the labelling is the other 8 and cannot be compressed by engineering.

E8 -- Data erasure (PDPL) -- 5 days (revised, finding B)

Exists: `dsr.service.ts` records access/erase requests as **pending** rows with audit events. **No actual deletion executes**, in Postgres or Neo4j.

Build:

- 1 **POST** `/api/v1/nscre/erase-patient` in `services/veritas-graph` --
`MATCH (p:Patient {id})-[:HAS_ENCOUNTER*0..1]->()-[r]->(n) DETACH DELETE`, scoped to patient-fact labels only. **Must not delete reference nodes** (`NphiesDrug`, `DoseRule`, ontology `Medication`) -- those are shared vocabulary. This is the one place to be careful; get the label allowlist explicit and test it.
- 1 An erase executor in `dsr.service.ts` that fans out to Postgres tables + the graph endpoint, transitions the request **pending** -> **completed**, and audits.
- 1 Tests: erasure removes all patient facts; **reference data survives**; idempotent re-run; a failed graph call leaves the request **pending**, never falsely **completed**.

Acceptance: an erase request provably removes the patient from both stores, with the graph reference vocabulary intact.

E5 (discovery) -- ICD-10-AM vs CM -- 1 day (engineering)

Spike only: confirm the required system URI, then size the change. Pre-scoped so the answer converts straight into work:

If the answer is	Change surface
ICD-10-AM (current)	None -- confirm and move on
ICD-10-CM	<code>app.snomed_icd10am_map</code> (rename + reseed), <code>icd-coding.service.ts</code> , the FHIR system URI in <code>connector.service.ts</code> and <code>config/nphies_profiles.json</code> , seeded dev data, <code>OrderLine.icd10Code</code> display, docs. ~5 days once the code set is in hand

NEXT -- 3 months

E3 -- Claim-integrity pilot package -- 15 days

The roadmap's fastest revenue path. Almost entirely assembly of built parts.

- **Deployment profile** that runs claim readiness, coding, rejection risk and pre-auth **without** the clinical agents -- a feature-flag/module composition in `app.module.ts`, so no SaMD surface ships. (~3d)
- **Rejection-cost dashboard** -- SAR at risk by department and code. Data exists in `rejection-risk.service.ts`; this is aggregation + a page. (~5d)
- **Claim simulator** -- "check before you send" against an existing claim batch; reuses `claim-readiness` + `validate_order_necessity`. (~4d)
- **Coder review queue** -- batch surface for RCM teams; the current flow is clinician-first only. (~3d)

E5 (execution) -- NPHIES conformance -- 8 days + external

Real IG profiles into `config/nphies_profiles.json` (already config-driven -- this was the Sprint 9 design decision paying off), conformance testing, and certificate lifecycle with annual renewal. Flip `profiles_verified` to `true` **only when genuinely verified**; the flag is surfaced on `/health` and should stay honest.

E9 -- ATC therapeutic-class data -- 4 days

Reference dataset + `(:Medication)-[:HAS_ATC_CLASS]->(:AtcClass)` edges, and a class filter in `screen_alternative_candidates()`. Removes the "Warfarin as a Metformin alternative" output and lets one of the two documented `limitations` be dropped.

Localisation quick wins -- 4 days

Hijri dates, Arabic patient-message templates, clinician-gender scheduling preference. Small, visible, repeatedly requested.

LATER

Item	Effort	Trigger
E6 -- durable queues (Celery/Redis or Postgres outbox)	8d	Second site or second replica
SMART-on-FHIR launch	15d	EMR embedding required by a buyer
E7 -- dialect ASR	25d+	Only if pilot shows ambient is the wedge
Ambient consent capture	3d	Before any real ambient use (PDPL)
SOC2/HITRUST	60d+	Non-KSA or large private group

Summary

Phase	Days	Calendar (1 eng)	Calendar (2 eng)
NOW (E2, E4, E1, E8, E5-discovery)	28	~6 weeks	~3.5 weeks
NEXT (E3, E5-exec, E9, localisation)	31	~7 weeks	~4 weeks

Phase	Days	Calendar (1 eng)	Calendar (2 eng)
LATER	111+	trigger-driven	--

The NOW phase fits the roadmap's 6-week window **for one engineer only if E1's labelling runs in parallel with a clinical resource**. With two engineers it lands in ~3.5 weeks -- and E1's dataset work is still the critical path, because it is not engineering-bound.

Highest leverage, in order: E2 (4d -- closes H-1, kills LLM lock-in, converts the residency gate to internal), E4a (2d -- removes the worst demo risk), then E8 (5d -- the compliance item with no current implementation at all).